STAT 7100 - Final Project Proposal

Project Title: U.S. Airline Performance Comparison and Forecast

1. Research Question & Motivation

I am aiming to split this project into 2 parts:

1. **Part 1:** To analyze and compare the historical delay performance of major airlines, identifying their long-term trends.
2. **Part 2:** To develop a macro-level forecast to predict the total volume of delays in future months, providing a systemic, forward-looking view.

Overall, my project direction is, what are the best and worst performing airlines in the US in terms of on-time flights, and what can we expect from them over the coming year.

Motivation: A simple "Top 10" list is a snapshot, but it doesn't show long-term accountability or improvement. The main goal is to use first identify the most impactful (both positively and negatively) airlines in the current US domestic travel market and use the statistical methods from this course to create a simple predictive forecast.

2. Dataset Description

Data Source & Citation: The primary dataset is the Airline_Delay_Cause.csv file (2003-2025), I got this dataset from Kaggle.com. It comes with the dataset itself and a short table explaining each variable.

LINK: https://www.kaggle.com/datasets/ryanjt/airline-delay-cause

It has **51** unique airlines, and **425** unique airports. All throughout the US. After cleaning, it has approximately 398,000 observations and 21 variables.

Data Cleaning & Key Variables:

The data is *mostly* clean, however there are some duplicates in airport_names, however that can be fixed and my main focus is on the performance of the **airlines** and not the airports.

1. The carrier_name column will be used as the unique airline identifier (per the data dictionary), as it represents each airline. The code variable is misleading because it does not account for **CODESHARE** flights, which is a common business practice.
2. A new variable, avg_delay_per_flight, will be added to the dataset by calculating arr_delay / arr_flights. This variable is going to be the main one for the analysis.

3. Analysis Plan

Part 1: Historical Trend Analysis for 2003-2025 (Simple Linear Regression)

> **Note:** I plan to use dplyr and ggplot2 on top of Base R to make the aggregation easier and generate some better looking plots than what Base R does. Also, this project will mostly use the carrier_name, airport_name, and arr_delay variables for all 3 parts.

1. **Aggregation**: The dataset will be aggregated to create a single, mean performance KPI for each airline for each year of operation.
2. **Regression:** A Simple Linear Regression model will be fit for each airline to model its performance trend over time.
3. **Ranking:** The slope of the regression line will be extracted from each model to serve as the quantitative measure of improvement (a negative slope) or regression (a positive slope). All airlines will be ranked by this metric.
4. **Visualising:** A summary plot will be created (using ggplot2) to present the Top 5 and Bottom 5 performing airlines

Part 2: Predictive Forecast (12-Month Horizon)

1. **Forecasting:** I will have10 simple linear regression models from Part 1 (for the Top 5 and Bottom 5 airlines). They will be used to generate a 12-month forecast. A new data frame representing the 12 future time periods will be created. The predict() function in R will be applied to each airline's model to calculate the expected performance for those 12 months.

2. **Prediction Intervals**: To determine the "reliable range" of future performance, 95% prediction intervals will be generated for the forecast. This will be done using the standard predict() function's built-in interval capabilities, which are based on the model's residual standard error. From that, I will have a "cone of uncertainty" around the main prediction.
3. **Visualising:** For this part, it will consist of two multi-panel charts from ggplot2, one for the Top 5 and one for the Bottom 5 airlines. These charts will be focused on the forecast. Each panel will display the airline's historical monthly performance data, the fitted simple linear regression line, the 12-month forecasted line, and the 95% prediction interval plotted as a shaded "cone of uncertainty."